

SOC Home Lab

Windows Security Monitoring & Brute-Force Detection

PROJECT DOCUMENTATION REPORT

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Career track: SOC Analyst (entry-level)

Environment: Splunk Enterprise 10.4.3 · Windows 11 (VMware)

Repository: github.com/dhirajantre/soc-home-lab-splunk-windows-monitoring

All authentication failures referenced in this report were intentionally generated in a controlled home-lab environment for detection-engineering practice. No production system or real attacker was involved.

Executive Summary

This report documents a self-directed SOC (Security Operations Center) home lab built to practice an end-to-end security-monitoring workflow on Splunk Enterprise. The lab collects Windows Security Event Logs, applies a custom SPL detection rule to identify brute-force authentication activity, fires a scheduled alert, and carries the resulting event through investigation and MITRE ATT&CK-mapped analyst assessment.

The detection logic was validated with a controlled test: three incorrect password attempts against a local Windows account within a five-minute window. Splunk correctly grouped the activity, triggered the configured alert, and the resulting event was investigated and mapped to MITRE ATT&CK technique T1110 (Brute Force), sub-technique T1110.001 (Password Guessing). The activity was classified as a controlled lab test, with no confirmed malicious intent.

Detection	Alert	Dashboard	Investigation	MITRE Mapping
Built & validated	Configured & fired	Live, 5 panels	Completed	T1110 / T1110.001

1. Project Overview

A hands-on SOC home lab built with Splunk Enterprise to collect, monitor, investigate, and detect Windows authentication activity. The project follows a complete SOC workflow: Windows Security Logs -> SIEM -> Detection Engineering -> Alerting -> Investigation -> MITRE ATT&CK Mapping -> Response.

Objectives

- Collect Windows Security Event Logs in Splunk
- Monitor authentication activity
- Investigate failed and successful logons
- Develop a brute-force detection rule
- Configure a scheduled security alert
- Validate the detection using controlled lab activity
- Map the detected behavior to MITRE ATT&CK
- Document investigation findings and response recommendations

Tools & Technologies

- Splunk Enterprise 10.4.3
- Windows 11 (VMware virtual machine)
- Windows Event Logs
- SPL (Splunk Search Processing Language)
- MITRE ATT&CK framework

Windows Events Investigated

Event ID	Description
4624	Successful Logon

Event ID	Description
4625	Failed Logon
4672	Special Privileges Assigned to New Logon

2. Detection Engineering

Detection Objective

Detect three or more failed Windows logon attempts for the same user and source within a five-minute window.

SPL Detection Query

```
index=* sourcetype=WinEventLog:Security EventCode=4625
| eval User_Account=mvfilter(Account_Name!="DHIRAJ-WIN11$")
| where isnotnull(User_Account)
| bin _time span=5m
| stats count as failed_attempts by _time User_Account Source_Network_Address
| where failed_attempts >= 3
| sort - failed_attempts
```

Detection Logic

- 1 Filters Windows failed authentication events using Event ID 4625
- 2 Extracts the user account, excluding the machine account
- 3 Groups events into five-minute windows
- 4 Groups activity by user and source network address
- 5 Counts failed authentication attempts per window
- 6 Returns a detection when the count reaches three or more

3. Alert Configuration

A scheduled Splunk alert operationalizes the detection so it runs continuously without manual searching.

Alert Name	Brute Force - 5 Minute Failed Login Detection
Alert Type	Scheduled
Schedule	Every 5 minutes (cron: */5 * * * *)
Time Range	Last 5 minutes
Trigger Condition	Number of Results > 0
Trigger Frequency	Once
Alert Expiration	24 hours
Severity	Medium
Action	Add to Triggered Alerts

Permissions

Private

splunk>Enterprise

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Modules

Alert title

Brute Force - 5 Minute Failed Login Detection

Description

Detects 3 or more failed Windows logon attempts for the same user and source within a 5-minute window.

Schedule & trigger conditions

Trigger actions

* Alert type

Scheduled

Real-time

* Schedule

Run

On Cron Schedule

Cron Expression

* / 5 * * * *

e.g. 0 18 * * * * (every day at 6PM). [Learn more](#)

* Time range

Last 5 minutes

Expires after

24

hour(s)

* Trigger alert when

Number of results

is greater than

0

* Trigger frequency

Once

For each result

Turn on throttle

☐

If throttle is on, then after an alert is triggered, subsequent alerts are suppressed for the duration of the throttle period. [Throttle alerts documentation](#)

Cancel

Save

1 of 1 pages

App

Sharing

Status

search

Private

Enabled

search

Private

Enabled

Figure 1. Scheduled alert configuration in Splunk, showing the cron schedule, time range, and trigger condition.

4. Detection Validation

The detection was validated with a controlled test: three incorrect passwords were intentionally entered for the test account. On its next five-minute cycle, the scheduled alert evaluated the search, found the qualifying activity, and fired successfully.

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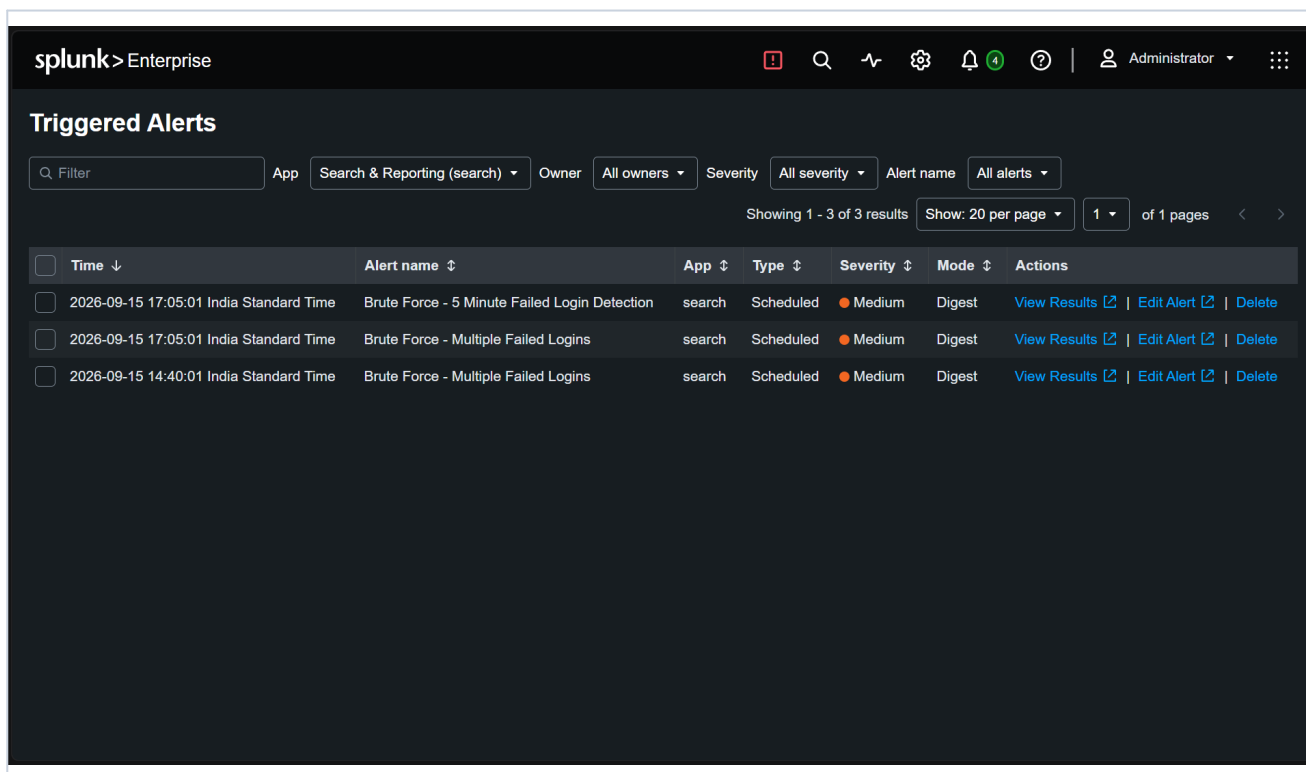


Figure 2. Triggered-alert history in Splunk, confirming the detection fired.

Two entries share the same detection logic under an earlier working name ("Brute Force - Multiple Failed Logins") used while the alert was being refined; the finalized alert is Brute Force - 5 Minute Failed Login Detection, shown in Figure 1.

Detection Result

User Account	Dhiraj
Source Address	127.0.0.1 (localhost)
Failed Attempts	3
Detection Window	5 minutes (17:00-17:05, 15 Sep 2026)
Event ID	4625
Logon Type	2 - Interactive

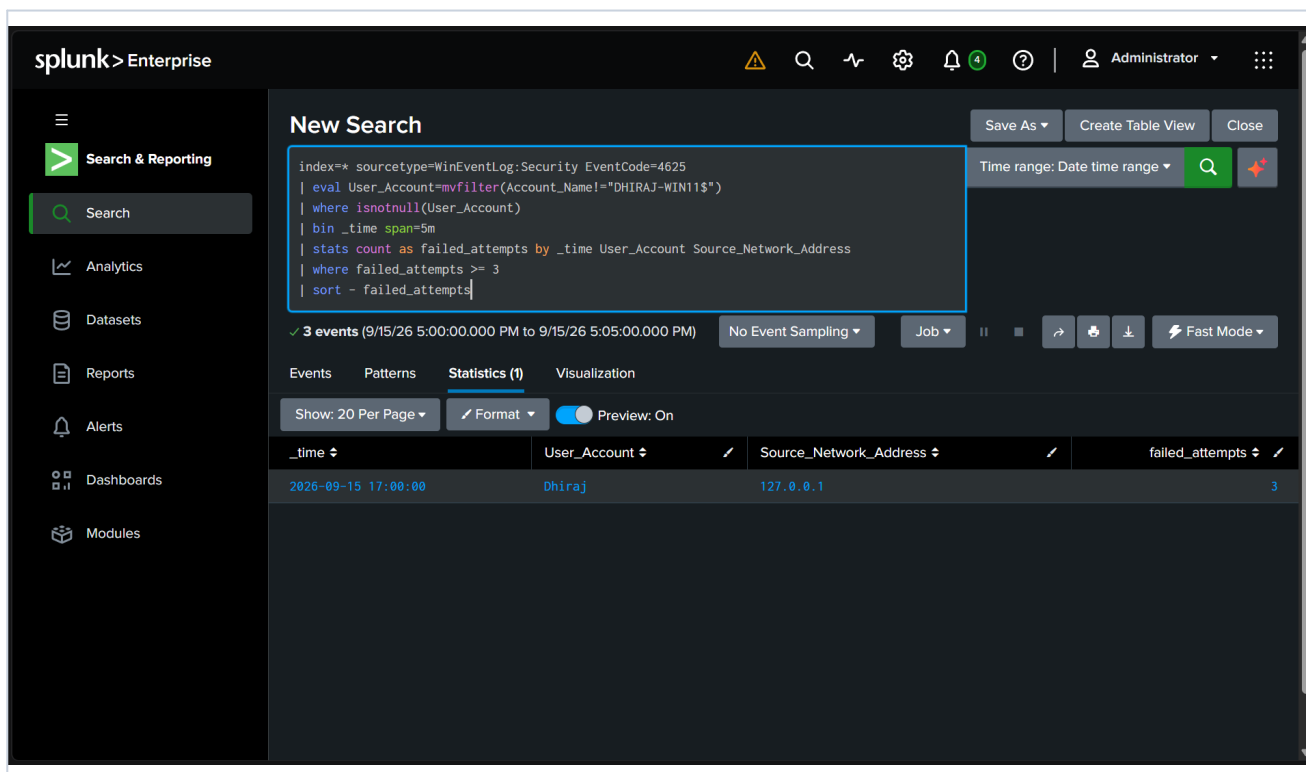


Figure 3. SPL search aggregating failed logons into 5-minute windows, isolating the qualifying 3-attempt detection.

5. Incident Investigation

Widening the search to the surrounding 24-hour period shows 11 failed logon attempts against the same account, arriving in four separate bursts. One burst - three attempts between 17:01:46 and 17:01:51 - met the brute-force threshold and is the event carried through investigation below.

Observed Activity

- Event ID: 4625
- User Account: Dhiraj
- Source Address: 127.0.0.1
- Workstation: DHIRAJ-WIN11
- Logon Type: 2 - Interactive
- Status: 0xC000006D
- Sub-Status: 0xC000006A
- Failure Reason: Unknown user name or bad password
- Attempts: 3 within a 5-minute window

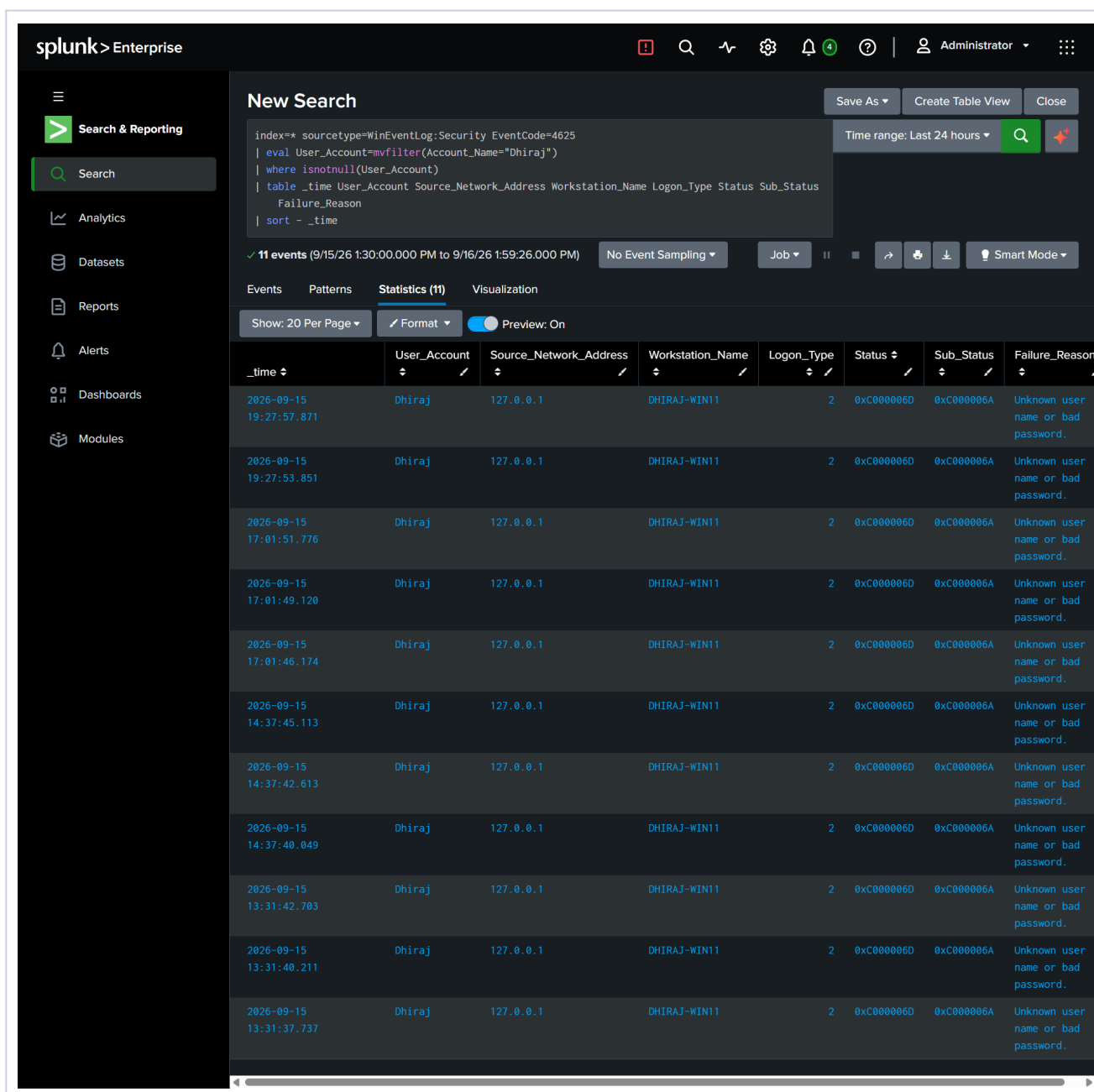


Figure 4. Full-day search of Event ID 4625 for the account, showing all 11 failed logons and their authentication fields.

Source Analysis

The source address for every attempt was 127.0.0.1, the localhost address of the system itself. The activity therefore shows no evidence of an external source in this controlled test.

Timeline

During the 17:00-17:05 detection window, three failed authentication attempts occurred for the same account and source. Splunk grouped the events and produced a single detection: User Account Dhiraj, Source 127.0.0.1, 3 failed attempts in a 5-minute window.

6. Analyst Assessment

The observed authentication pattern is consistent with password-guessing behavior: multiple failed attempts occurred for the same account within a short time window. However, the activity originated from 127.0.0.1

(localhost) and was intentionally generated as part of a controlled SOC laboratory test.

FINAL CLASSIFICATION: CONTROLLED LAB TEST -- NO CONFIRMED MALICIOUS ACTIVITY

No confirmed external attack or account compromise was identified from this test.

7. MITRE ATT&CK Mapping

Tactic	Technique	Sub-technique
Credential Access	T1110 - Brute Force	T1110.001 - Password Guessing

The repeated failed-authentication pattern observed during the controlled test is consistent with password-guessing behavior as defined by this sub-technique.

8. Recommended SOC Response

If similar activity were detected in a production environment, a SOC analyst could:

- 1 Validate whether the authentication attempts were authorized.
- 2 Review successful logon events following the failed attempts.
- 3 Investigate the source IP address and affected account.
- 4 Check for additional authentication anomalies.
- 5 Review related Windows Security events.
- 6 If malicious activity is confirmed, consider temporarily locking or disabling the affected account.
- 7 Continue monitoring the account and source for additional suspicious activity.

9. Splunk Dashboard

A Splunk Dashboard Studio dashboard, **SOC Home Lab - Security Monitoring**, provides a centralized, continuously updating view of Windows authentication activity.

Dashboard Panels

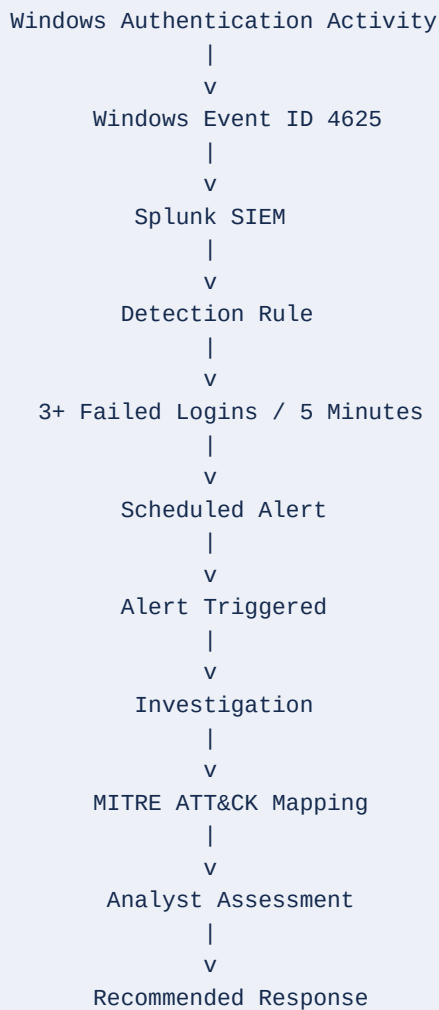
- Failed Windows Logins
- Successful Windows Logins
- Brute Force Detections
- Privileged Logons
- Failed Login Activity Over Time (hourly trend)



Figure 5. SOC Home Lab dashboard, showing a rolling 24-hour view of Windows authentication activity.

Panel counts are a live, rolling 24-hour window rather than a historical log: at the time of this capture the validated 17:00-17:05 detection had aged out of the window, so Brute Force Detections reads 0. Figures 2 and 3 confirm the detection fired correctly when the activity occurred.

10. SOC Workflow



11. Skills Demonstrated

- SIEM & Security Monitoring
- Windows Event Log Analysis
- SPL (Splunk Search Processing Language)
- Detection Engineering & Alert Triage
- Authentication Monitoring
- Incident Investigation & Source/IP Analysis
- MITRE ATT&CK Mapping
- SOC Workflow & Incident Documentation

Key Learning Outcomes

This project reinforced practical skills in:

- Collecting and analyzing Windows Security logs
- Understanding Windows authentication event IDs
- Writing SPL detection logic and time-based detection rules

- Configuring and validating scheduled security alerts
- Investigating authentication failures end to end
- Mapping observed behavior to MITRE ATT&CK
- Documenting findings using a structured SOC investigation workflow

Lab Disclaimer: All authentication failures in this project were intentionally generated in a controlled home-lab environment for cybersecurity learning and detection validation. The observed activity should not be interpreted as a confirmed real-world attack.

12. Project Status: Completed

[x] Windows Security Log Collection	[x] Authentication Event Analysis
[x] Splunk SIEM Configuration	[x] Detection Engineering
[x] Splunk Dashboard	[x] Scheduled Alert
[x] Alert Trigger Validation	[x] Incident Investigation
[x] MITRE ATT&CK Mapping	[x] SOC Response Recommendations
[x] Project Documentation	[x] Evidence Screenshots

13. Conclusion

This home lab demonstrates a complete, self-directed SOC workflow: Windows log collection, SPL-based detection engineering, scheduled alerting, incident investigation, and MITRE ATT&CK-mapped analyst assessment, built and validated end to end on Splunk Enterprise.

Full project files, including the SPL queries, alert configuration, and evidence screenshots, are available in the public repository below.

github.com/dhirajantre/soc-home-lab-splunk-windows-monitoring